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RESEARCH ARTICLE

Personalized Nutrition-Aware Dietary Recommendation With Multimodal Large Language Models

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ABSTRACT Poor diet quality is a major modifiable risk factor for obesity, cardiovascular disease, and metabolic disorders, motivating a growing ecosystem of diet-tracking apps and recipe platforms that promise personalized guidance. However, most of these systems still rely on simple scoring rules or ID-based collaborative filtering that ignore rich multimodal food information (ingredients, cooking methods, images, nutrient profiles) and lack explicit nutritional reasoning. To address this gap, we propose *NutriRec*, a multimodal LLM-based framework for personalized dietary recommendation that links nutritional reasoning with collaborative dietary patterns. We uses a multimodal LLM to generate unified representations of meals and food items from textual descriptions, ingredient lists, nutrient profiles, and images, capturing both sensory and nutritional attributes. On top of these content-based embeddings, NutriRec incorporates collaborative filtering signals through a post-alignment mechanism that aligns user representations from an ID-based sequential model with those from the multimodal LLM. This design enables explainable dietary reasoning by grounding each recommendation in concrete nutritional evidence (e.g., lower sodium, higher fiber, heart-healthy tags) while still reflecting behavioral patterns learned from large-scale consumption logs. We evaluate NutriRec on realistic dietary recommendation scenarios with long-term, multimodal food interaction histories. Across these settings, NutriRec not only yields stronger recommendations than sequential and LLM-based baselines, but also provides more explainable dietary reasoning by explicitly linking suggestions to nutrient profiles, meal composition, and simple guideline-based constraints.

INDEX TERMS Personalized nutrition, dietary recommendation, large language models, multi-modal recommendation.

I. INTRODUCTION

Diet-related chronic conditions such as obesity, type 2 diabetes, and cardiovascular disease remain major contributors to global morbidity and mortality [1]. A long line of epidemiological work shows that it is usually whole dietary patterns, rather than single nutrients in isolation, that drive long-term health outcomes [1]. In recent years, diet-tracking applications, online recipe platforms, and grocery delivery services have created new channels through which individuals receive day-to-day dietary advice at scale [2], [3], [4]. Yet many of these tools still behave as simple calorie counters or generic recommender systems that prioritize clicks and purchases, with only limited attention to nutritional quality or clinical constraints [5], [6].

On the algorithmic side, food and grocery platforms commonly rely on collaborative filtering and sequential recommendation techniques that have been successful in other consumer domains [7], [8], [9], [10]. These ID-based models [11], [12] treat each recipe or product as an abstract item and learn latent representations from historical co-consumption sequences [13], [14]. While such approaches capture regularities in what people tend to eat together or over time, they are largely blind to nutritional content [15], [16]: they do not distinguish a high-sodium from a low-sodium dinner, recognise whether a dish fits a diabetic-friendly pattern, or consider how a sequence of meals accumulates energy and sodium above recommended thresholds. Recent

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food recommender studies have begun to add basic nutrient features or health scores as side information [17], [18], [19], but in most cases these variables are used as static filters or weak auxiliary signals, rather than being placed at the centre of the modelling and evaluation process.

At the same time, pre-trained language and vision models make it possible to work directly with the raw artefacts of eating: recipe titles, ingredient lists, cooking instructions, dish images, and structured nutrient tables [20], [21], [22]. For dietary applications, this means that models no longer have to rely only on abstract IDs or a small set of hand-crafted features; they can, in principle, read and interpret the same information that human dietitians use. However, if these rich representations are used in isolation, without reference to large-scale consumption logs, they risk producing recommendations that are nutritionally sound on paper but poorly matched to what users actually choose in daily life [5], [18], [23]. From a public health perspective, an effective system must therefore bring together three elements: realistic behavioural patterns, detailed multimodal information about foods, and transparent reasoning about why a given suggestion is better or worse than a user's typical choices.

In this work, we introduce NutriRec, a multimodal framework designed for personalized dietary recommendation that integrates nutritional reasoning with collaborative dietary patterns. NutriRec consists of two complementary components. First, a multimodal nutritional item representation model encodes meals and food products by incorporating recipe text, ingredient lists, structured nutrient profiles, and images [21], [22]. This approach generates embeddings that explicitly capture both concrete nutritional attributes (e.g., energy density, sodium, fibre) and sensory cues (e.g., cuisine type, cooking method). Second, a dynamic user dietary profile generator aggregates these item embeddings based on a user's meal or purchase history, creating a content-based representation of their evolving dietary patterns.

In addition to the content-based component, NutriRec also retains collaborative information derived from large-scale behavioral data. We train an ID-based sequential model on user–item interaction sequences, which captures regularities such as co-occurrence of meals within specific timeframes, common product substitutions, and the variations in dietary patterns across users with different preferences and goals [9], [24]. A post-alignment mechanism then synchronizes the content-based and ID-based user representations, ensuring that each recommendation reflects both behavioral patterns and tangible nutritional changes. For example, this approach can suggest meals that lower sodium relative to a user's recent dinners or increase fibre while maintaining alignment with familiar cuisines.

We evaluate NutriRec on three datasets—MealRec, foodRecSys, and an Amazon Food subset [25], [26]—which together cover home-cooked meals, community recipe platforms, and grocery shopping logs. Beyond standard ranking metrics, we also assess nutrition-aware performance through case-based evaluations that focus on meal-level health scores, adherence to simple dietary guidelines, and the interpretability of dietary recommendations [5], [19]. Our contributions are threefold:

- We define nutrition-aware dietary recommendation as a multimodal sequence task and develop NutriRec, which learns nutritionally informed meal representations from text, ingredients, images, and nutrient data, creating dynamic user profiles that provide transparent explanations of meal composition and nutrient balance.
- We propose a post-alignment mechanism that aligns content-based and ID-based user representations, enabling NutriRec to reflect realistic dietary patterns while suggesting healthier alternatives in an interpretable way.
- We perform extensive experiments on MealRec, foodRecSys, and an Amazon Food subset, showing consistent improvements over strong baselines in both recommendation quality and nutrition-aware metrics, focusing on guideline-compliant dietary improvements.

II. RELATED WORK

A. FOOD AND NUTRITION RECOMMENDER SYSTEMS

Early food recommenders largely transplanted techniques from generic collaborative filtering and content-based recommendation, matching users and items by flavour similarity, cuisine type, or overlapping ingredients [27], [28]. More recent surveys map out the broader landscape of food and grocery recommenders and highlight their potential, as well as their limitations from a public health perspective [18], [19]. In most deployed systems, the core signal remains user–recipe or user–product interactions, occasionally enriched with high-level tags such as ''vegetarian'' or ''low-carb'', while explicit nutrient constraints or health scores appear only in a subset of studies [17], [29]. When present, these nutrition signals are typically treated as static side information or post-hoc filters rather than being integrated into the main modelling and evaluation pipeline.

On the nutrition side, there is a growing body of work linking recommender techniques with personalized nutrition and precision health. Qiao et al. argue for food recommendation that explicitly balances enjoyment, feasibility, and health outcomes [5]. Parallel review papers discuss the use of digital tools and AI methods for individualized nutrition, drawing attention to issues such as data quality, long-term adherence, and clinical validation [2], [3], [4]. Guo et al. further review ChatGPT-based meal planning and dietary counselling, and conclude that generic models can generate plausible plans but need to be anchored in domain-specific constraints [6].

B. SEQUENTIAL AND MULTIMODAL RECOMMENDATION

Sequential recommendation has been extensively studied as a way to capture the temporal dynamics of user behaviour [10]. Representative approaches range from Markov chain models for next-basket prediction [9], to RNN-based session models [30], [31], and more recent self-attention architectures such as SASRec and BERT4Rec [13], [14]. Hybrid architectures that incorporate convolutional or

contrastive components further improve robustness and generalisation [24], [32], [33], [34]. These models underpin many large-scale e-commerce and media platforms, but they tend to treat item IDs as atomic symbols and rely on co-occurrence patterns to encode meaning, with little direct access to the semantic or nutritional content of items.

To address this, multimodal recommenders integrate textual descriptions, images, and other side information into the user–item modelling process. Examples include multimodal graph convolutional networks for micro-video recommendation [35], [36], self-supervised multimodal representation learning [37], and hierarchical contrastive frameworks such as MENTOR [38]. These studies demonstrate that aligning visual and textual modalities can substantially improve performance, particularly in cold-start or sparse-feedback settings. Our work follows this multimodal line but differs in two key aspects: we explicitly introduce structured nutrient vectors as an additional modality carrying normative health information, and we use a multimodal language–vision backbone to place text, images, and nutrients in a unified representation space tailored to dietary decision-making.

C. LARGE LANGUAGE MODELS FOR RECOMMENDATION AND PERSONALIZED NUTRITION

The rapid development of large language models has prompted a range of studies on their use in recommendation. Some works recast recommendation as an instruction-following or conversational task, where models generate ranked lists directly from textualised user histories [39], [40], [41]. Others use pretrained language representations to enrich user and item profiles or to provide knowledge augmentation in open-world settings [42], [43], [44]. A complementary line seeks to connect language-based representations with collaborative filtering signals: CollM and LLara inject ID-based embeddings into language models or align them during training [45], [46], while LLMRec and related approaches use language models to augment interaction graphs or denoise sequential signals [47], [48], [49]. Recent evidence suggests that frozen language representations can themselves encode strong collaborative structure, inspiring models such as AlphaRec and IRLLRec that build recommenders directly on top of language-derived item or intent embeddings [50], [51], [52].

In parallel, the nutrition and dietetics community has begun to explore large language models as tools for dietary counselling and meal planning. Systematic reviews report that systems such as ChatGPT can answer common nutrition questions and compose meal plans, but often do not consistently follow dietary guidelines or adapt to individual health conditions [6], [53]. These observations point to the need for models that are grounded in nutrient profiles, health scores, and longitudinal dietary logs, rather than operating purely at the level of generic text.

NutriRec sits at the intersection of these strands. Methodologically, it borrows the idea of decoupling content-based and collaborative modelling from recent alignment frameworks [45], [47], [48], but instantiates it in a nutrition-centred setting with multimodal meal representations and explicit health objectives. Conceptually, it extends AI-for-nutrition work by showing how detailed multimodal descriptions of food and population-level dietary patterns can be combined to produce recommendations that are not only accurate, but also interpretable in nutritional terms.

III. METHODS

This section presents the framework and technical design of *NutriRec*, a multimodal dietary recommendation system that integrates nutritional reasoning with collaborative dietary patterns. NutriRec utilizes a hybrid approach that combines multimodal item representations, sequential user behavior modeling, and collaborative filtering signals. The system's key components are outlined in the following sections.

IV. METHODS

We present *NutriRec*, a framework designed to bridge the gap between nutritional reasoning and collaborative filtering. Unlike systems that rely solely on latent IDs or isolated content features, NutriRec synergizes multimodal item encoding with sequential behavior modeling. The architecture consists of three distinct yet interconnected modules: a Multimodal Nutritional Item Representation model that fuses sensory and nutrient data; a Dynamic User Dietary Profile Generator that tracks evolving content preferences; and a Sequential User Behavior Model that captures implicit collaborative patterns.

A. MULTIMODAL NUTRITIONAL ITEM REPRESENTATION

Central to *NutriRec* is the extraction of high-fidelity item representations that transcend simple identifiers. To accurately model the multifaceted nature of food choices, we construct a unified embedding space that synthesizes sensory appeal with nutritional facts. As illustrated in Figure 2, our encoder processes three complementary modalities:

- **Recipe and Ingredient Texts:** We leverage descriptive texts—spanning ingredient lists, preparation steps, and flavor descriptors—to capture the semantic granularity of a meal. This modality encodes critical sensory aspects, such as cooking techniques (e.g., ''grilled'' vs. ''fried'') and flavor profiles, which are pivotal for understanding user taste preferences and dietary compatibility.
- **Meal Images:** Visual aesthetics are a primary driver of food selection. We utilize meal images to encode non-textual sensory signals, including plating style, texture, and color vibrancy. These visual embeddings provide context that complements the textual descriptions, offering a more holistic representation of the item's sensory appeal.
- **Nutrient Profiles:** To ground recommendations in biological reality, we incorporate structured macro- and micro-nutrient data (e.g., protein, sodium, fiber). Recognizing that real-world data is often sparse, our framework employs an adaptive encoding strategy: detailed nutrient vectors are integrated when available to

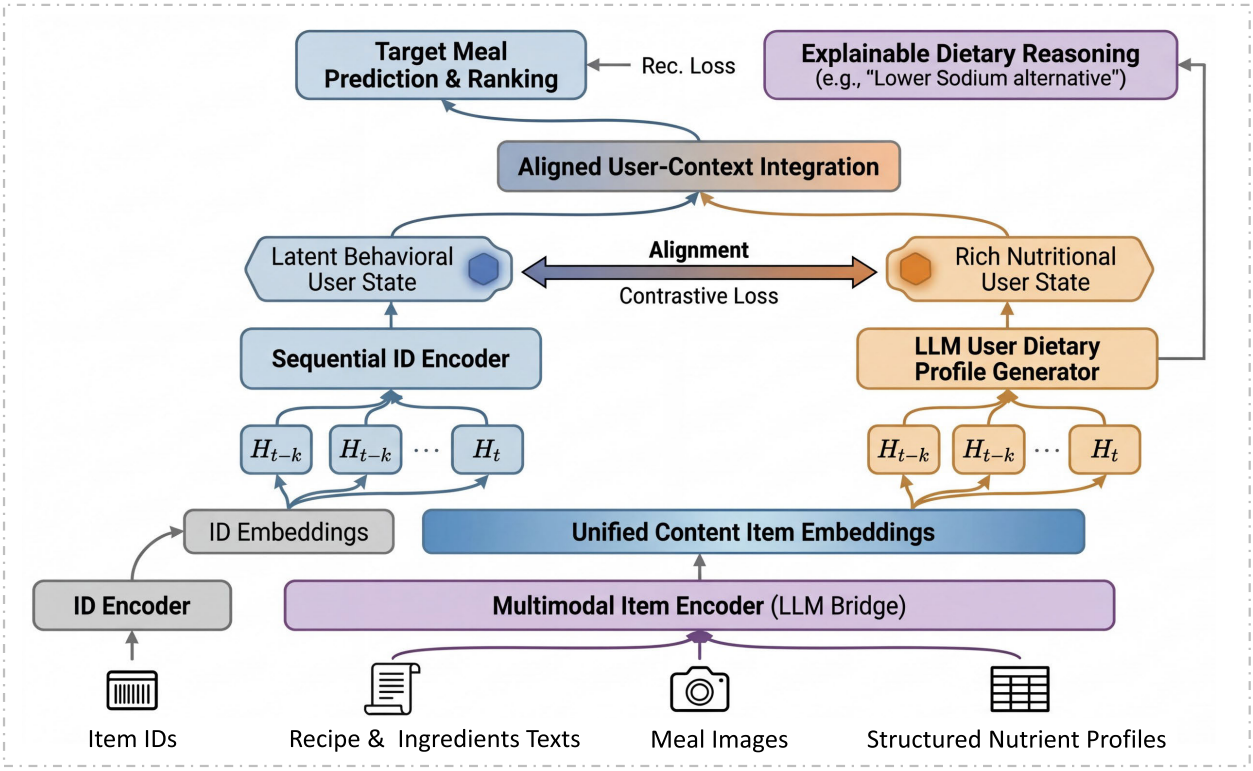


FIGURE 1. Overview of NutriRec. A multimodal item encoder at the bottom (LLM bridge) fuses recipe and ingredient texts, meal images, and structured nutrient profiles into unified content item embeddings, alongside an ID encoder that produces item-ID embeddings. A sequential ID encoder (left) aggregates interaction sequences into a latent behavioural user state, while an LLM-based user dietary profile generator (right) summarizes content item embeddings into a rich nutritional user state. An alignment module with contrastive loss ties these two user states together, and the aligned user–context representation is used for target meal prediction and ranking as well as explainable dietary reasoning (e.g., suggesting a lower-sodium alternative).

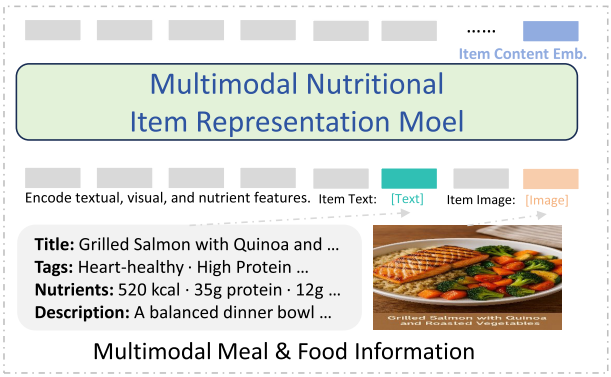


FIGURE 2. Overview of the Multimodal Nutritional Item Representation model. The module fuses heterogeneous data sources—including unstructured recipe texts, visual content, and structured nutrient profiles—into a unified latent embedding. This design captures the interplay between sensory attributes and nutritional composition, facilitating recommendations that balance user preference with dietary healthiness.

enforce precise health constraints, while the model falls back on text-inferred attributes in their absence.

Technically, these inputs are fused via a multimodal encoder. We frame the processing of recipe and ingredient texts as a prompt-based task for a Large Language Model (LLM), extracting dense semantic representations. Concurrently, visual features are derived via a vision encoder. These modalities are projected into a shared latent space, augmenting the representation with structured nutrient vectors where applicable. This fusion ensures that the resulting embedding encodes not just *what* the item is, but *how* it aligns with both sensory cravings and nutritional standards.

B. DYNAMIC USER DIETARY PROFILE GENERATOR

Capturing the evolution of user preferences requires more than static profiling. *NutriRec* employs a Dynamic User Dietary Profile Generator to construct a time-aware content representation of the user. Instead of relying on static tags, this component recursively aggregates the multimodal item embeddings from a user's historical consumption sequence.

By processing the sequence of consumed meals, the model distills high-level dietary patterns—such as a predilection for specific ingredient combinations, adherence to cuisine types, or consistent nutrient restrictions (e.g., low-sodium choices). As the user interacts with new items, this profile is dynamically updated, ensuring the representation remains synchronized with shifting tastes and health goals. This continuous aggregation allows the system to model the trajectory of a user's dietary habits, moving beyond immediate clicks to understand long-term nutritional behavior.

C. SEQUENTIAL USER BEHAVIOR MODELING

While content-based profiling captures *explicit* preferences and nutritional needs, it may miss the *implicit* signals embedded in community behavior. To address this, *NutriRec* runs a parallel Sequential User Behavior Model based on collaborative filtering. This ID-based branch learns from large-scale interaction logs to capture latent behavioral dynamics that content features alone might overlook.

This module encodes the temporal dependencies in user-item interactions, identifying patterns such as substitution logic (e.g., swapping similar items), complementary pairing, and seasonal trends. By aggregating these interactions into a latent behavioral state, the model ensures that recommendations are not only nutritionally sound but also anchored in realistic consumption distributions. This dual-track approach—combining the explicit nutritional profile with implicit behavioral signals—enables *NutriRec* to generate recommendations that are both biologically beneficial and behaviorally plausible.

D. LOSS FUNCTIONS AND ALIGNMENT MECHANISM

To ensure robust item retrieval while preserving semantic consistency, we optimize *NutriRec* using a joint optimization framework. This objective combines a recommendation loss, which maximizes the likelihood of observed user-item interactions, with a contrastive alignment loss, which serves as a regularizer to synchronize the latent spaces of the collaborative and multimodal modules.

1) POINT-WISE RECOMMENDATION LOSS

We formulate the next-item prediction task as a binary classification problem. To model implicit feedback effectively, we employ the Binary Cross-Entropy (BCE) loss. This objective encourages the model to assign higher confidence scores to ground-truth interactions (positive samples) while suppressing the scores of unobserved items (negative samples). We adopt a 1:1 negative sampling strategy, pairing each observed interaction with a randomly sampled non-interacting item to approximate the full softmax distribution efficiently.

Let $y_{ui} \in \{0, 1\}$ denote the ground-truth label indicating whether user u interacted with item i, and let $\hat{y}_{ui}$ be the predicted probability (logit after sigmoid activation). The recommendation loss is defined as:

$$\mathcal{L}_{bce} = -\sum_{(u,i)\in\mathcal{D}} \left(y_{ui} \cdot \log(\hat{y}_{ui}) + (1-y_{ui}) \cdot \log(1-\hat{y}_{ui})\right) \quad (1)$$

Minimizing $\mathcal{L}_{bce}$ optimizes the model's ability to discriminate between relevant and irrelevant meals based on historical behavioral patterns.

2) ALIGNMENT LOSS

Solely optimizing for interaction likelihood can lead to overfitting on ID-based patterns, neglecting the rich nutritional semantics encoded by the LLM. To bridge the representational gap between the collaborative space (ID-based) and the semantic space (Content-based), we introduce a contrastive alignment objective. This mechanism enforces that the *Sequential User Behavior* representation (E_u^{id}) and the *Dynamic Dietary Profile* (E_u^{con}) for the same user are mutually grounded in the same latent manifold.

We adopt a symmetric InfoNCE loss to maximize the mutual information between the two views. Formally, for a batch of users $\mathcal{U}$, we treat (E_u^{id}, E_u^{con}) as a positive pair, while treating the embeddings of other users $v \in \mathcal{U} \setminus \{u\}$ as negative samples. The alignment loss is computed as:

$$\mathcal{L}_{align} = -\frac{1}{|U|}\sum_{u=1}^{|U|}\left(\begin{array}{l}\log \frac{\exp(s(E_u^{id}, E_u^{con})/\tau)}{\sum_{j\in K}\exp(s(E_u^{id}, E_j^{con})/\tau)} \\ +\log \frac{\exp(s(E_u^{con}, E_u^{id})/\tau)}{\sum_{j\in K}\exp(s(E_u^{con}, E_j^{id})/\tau)}\end{array}\right) \quad (2)$$

where $s(\cdot, \cdot)$ denotes the cosine similarity, and τ is the temperature parameter controlling the sharpness of the distribution. By minimizing $\mathcal{L}_{align}$, we pull the collaborative and nutritional representations of the same user closer while pushing apart distinct users, ensuring that the behavioral model remains semantically consistent with the nutritional profile generated by the LLM.

3) JOINT OBJECTIVE FUNCTION

The final training objective is a weighted summation of the recommendation and alignment losses. This multi-task learning formulation allows *NutriRec* to simultaneously learn accurate ranking policies and biologically grounded user representations. The total loss is defined as:

$$\mathcal{L}_{total} = \mathcal{L}_{bce} + \lambda \cdot \mathcal{L}_{align} \quad (3)$$

where λ is a hyperparameter that balances the contribution of the semantic alignment task. This joint optimization ensures that the system provides recommendations that are not only behaviorally plausible but also interpretable and aligned with the user's nutritional needs.

E. EXPLAINABLE DIETARY REASONING

Beyond accurate ranking, a critical objective of *NutriRec* is to bridge the gap between latent collaborative signals and human-interpretable rationale. While traditional ID-based models function as black boxes—maximizing likelihood without semantic context—our framework leverages aligned multimodal representations to generate transparent justifications. Specifically, the reasoning module articulates how a recommended item balances the user's historical preferences (e.g., cuisine, flavor profile) with necessary nutritional interventions (e.g., sodium reduction).

This process is operationalized by treating recommendation explanation as a conditional generation task. Once the aligned ranking model selects a candidate item i^* that maximizes the joint probability of user interest and nutritional compliance, we employ a frozen Large Language Model (e.g., Qwen2.5) as a reasoning engine. The LLM is prompted to contextualize the recommendation by contrasting the nutritional attributes of i^* against the user's high-risk consumption patterns.

To ensure the generated explanations are both factually grounded and structurally consistent, we design a schema-constrained prompt, denoted as T_{EDR}. As visualized in Figure 3, this prompt directs the LLM to quantify the "Nutrient Shift" and explicitly identify "Preference Retention," ensuring the output is not merely a generic description but a personalized comparative analysis.

Prompt Template: Explainable Dietary Reasoning (T_{EDR})

[Role Definition]
You are an expert AI Dietitian. Your task is to justify a recommendation that improves the user's health without sacrificing their taste preferences.

[Input Context]
`<User History>`: List of recent high-frequency meals with key nutrients (e.g., high sodium, fried).
`<Inferred Profile>`: e.g., "Prefers Spicy/Savory; High Sodium Risk."
`<Candidate Item>`: The healthy item recommended by the NutriRec model.

[Instruction]
Analyze the gap between `<User History>` and `<Candidate Item>`. Generate a structured justification containing:

- **Nutrient Shift**: Quantify the improvement (e.g., "Reduces sodium by X mg").
- **Preference Retention**: Explain how the item still fits the user's taste (e.g., "Retains spicy flavor using chili instead of salt").
- **Health Impact**: Link the change to a specific health benefit.

[Output Format]
Provide a JSON object with keys: `"nutrient_shift"`, `"preference_alignment"`, and `"natural_language_explanation"`.

FIGURE 3. The schema-constrained prompt (T_{EDR}) used for generating explainable reasoning. It explicitly instructs the LLM to analyze the trade-off between health improvement and preference retention.

By decoupling the reasoning stage from the ranking stage, *NutriRec* ensures that the high computational cost of generation is incurred only for the final recommended items, providing a scalable solution for explainable precision nutrition.

F. TRAINING AND INFERENCE

The operational workflow of *NutriRec* follows a decoupled paradigm, distinguishing between representation learning during training and semantic reasoning during inference.

1) TRAINING PHASE

To balance performance with computational efficiency, we adopt a parameter-efficient tuning strategy. The backbone Large Language Model (LLM) serves as a frozen semantic feature extractor, ensuring that the pre-trained knowledge base remains intact while minimizing memory overhead. Optimization is exclusively targeted at the lightweight mapping layers and the sequential user modeling components. Specifically, the mapping layers project multimodal inputs—spanning textual descriptions, visual cues, and structured nutrient profiles—into a unified latent space. Concurrently, the *Sequential User Behavior Model* and the *Dynamic User Dietary Profile Generator* are trained to synthesize these embeddings into evolving user state representations, capturing the synergy between historical interactions and nutritional preferences.

2) INFERENCE PHASE

The inference process extends beyond standard item ranking. Once the model retrieves candidate items based on the learned user profiles, the *Explainable Dietary Reasoning* module is invoked. Leveraging the aligned user-item representations, this module generates context-aware natural language justifications. This step explicitly grounds the recommendation in the user's specific health goals and flavor preferences (e.g., articulating why a lower-sodium variant was chosen over a historical favorite), thereby fostering user trust and actionable insight without affecting the latency of the initial retrieval stage.

V. EXPERIMENTAL SETUP

We focus on the following four research questions to comprehensively validate the effectiveness of **NutriRec**:

- **RQ1 (Overall Performance).** How does NutriRec compare against state-of-the-art ID-based, modality-based, and LLM-enhanced recommendation baselines in terms of general ranking accuracy across different food domains?
- **RQ2 (Nutritional Effectiveness).** Beyond standard ranking accuracy, can NutriRec effectively guide users toward healthier dietary patterns (e.g., improved health scores, guideline adherence) compared to baselines that rely solely on historical behavioral correlations?
- **RQ3 (Ablation & Model Design).** What is the contribution of the specific multimodal components (Recipe Text, Meal Images, Nutrient Profiles) and the alignment mechanism? How do key hyperparameters (e.g., the alignment weight α) impact the model's ability to balance preference matching with nutritional goals?
- **RQ4 (Explainability).** Does the Explainable Dietary Reasoning module generate biologically grounded and contextually relevant justifications that help users understand the nutritional value of the recommended meals?

A. DATASETS

We evaluate on three representative food and nutrition corpora that jointly cover structured meal planning, recipe sharing communities, and e-commerce food consumption. All experiments are personalized and use strictly chronological (log-order) splits to avoid future information leakage. To ensure data quality and reliable preference modeling,

TABLE 1. Datasets and nutrient–context signals. "Txt" lists the dominant platform–native narratives used to extract health and flavor insights. We apply a k-core filter to ensure data quality, removing users and items with insufficient interactions.

Dataset	#Users	#Items	#Bundles	#Interacts	Txt (nutrient–preference)
MealRec	1,575	3,826	987	151,148	Meal course descriptions (appetizer, main, dessert), nutritional tags, and healthiness scores.
FoodRecSys-V1	22,714	11,352	–	263,044	Recipe steps, ingredient lists, cooking directions, and user comments on taste/difficulty.
Amazon Foods	20,458	10,247	–	142,566	Product titles, detailed flavor descriptions, ingredient labels, and user review snippets.

we filtered out incomplete data entries and users/items with fewer than 5 interactions (5-core setting) across all datasets.

MealRec [25] is a comprehensive dataset designed for meal recommendation. It uniquely contains meal–course affiliations, linking users not just to individual items but to complete meal bundles (e.g., appetizer, main dish, dessert). We utilize the healthiness scores and course descriptions to extract nutrient-aware contexts.

FoodRecSys-V1 [26] is collected from Allrecipes.com, one of the largest food-oriented social networks. It provides rich textual modalities including ingredient lists, cooking directions, and user reviews. We focus on the recipe texts and user comments to derive insights about flavor profiles and preparation complexity.

Amazon Fine Foods [54] (subset) represents a large-scale e-commerce scenario. We construct a dense subset from the "Grocery and Gourmet Food" category. The text data here is characterized by product descriptions and explicit user feedback on taste and quality, which we process to capture fine-grained user preferences for specific ingredients and brands.

B. EVALUATION METRICS

We evaluate ranking quality with Hit Rate (HR) and Normalized Discounted Cumulative Gain (NDCG) at cutoff K. For a user u with ground-truth next item $i^\star$ and a ranked list π_u:

$$\mathrm{HR@}K(u) = \mathbb{1}\left\{i^\star \in \pi_u^{(K)}\right\}, \tag{4}$$

$$\mathrm{NDCG@}K(u) = \frac{\mathbb{1}\left\{\mathrm{rank}_{\pi_u}(i^\star) \le K\right\}}{\log_2\left(1+\mathrm{rank}_{\pi_u}(i^\star)\right)}. \tag{5}$$

We report the mean HR@K and NDCG@K over all test users. Unless otherwise stated, $K \in \{10, 20, 50\}$.

C. BASELINES

To demonstrate the effectiveness of **LLM-SCRec** (and the **NutriRec** framework), we compare it against three categories of state-of-the-art methods, ranging from classic ID-based models to advanced modality-aware and LLM-enhanced approaches.

Traditional Sequential Recommendation (ID-based). These methods rely solely on item ID sequences to model user preferences.

- **GRU4Rec** [55]: A seminal session-based recommendation model that applies Gated Recurrent Units (GRU) to capture sequential dependencies in user interactions.
- **SASRec** [13]: A self-attentive sequential recommendation model that utilizes a multi-head attention mechanism to identify relevant items from history, serving as a strong baseline for sequential modeling.
- **Caser** [32]: A convolutional sequence embedding model that treats user interaction history as an image, employing horizontal and vertical convolutional filters to capture high-order sequential patterns and skip behaviors.

Modality-based Sequential Recommendation (MoRec). These methods replace ID embeddings with modality encoders (e.g., text or image encoders) to leverage raw content features, addressing the cold-start problem and enhancing semantic understanding.

- **MoRec** [56]: A universal framework (also known as the "Where to Go Next" paradigm) that replaces the ID embedding layer of SASRec with end-to-end learned modality encoders (e.g., BERT for text), explicitly validating the effectiveness of pure modality-based representations.
- **UniSRec** [57]: A method designed for universal sequence representation learning. It utilizes item description text to learn transferable representations across different domains via a whitening-based contrastive objective.
- **VQRec** [58]: A state-of-the-art approach that maps item text/images to discrete code indices using Vector Quantization (VQ). It effectively handles the "long-tail" distribution of text features and filters noise in modality data.

LLM-based Recommendation. These methods leverage the reasoning and world knowledge of Large Language Models to enhance recommendation performance.

- **TALLRec** [41]: An instruction-tuning framework that formulates the recommendation task as a prompt-based generation problem, fine-tuning LLaMA with user interaction history instructions.
- **LlaRA** [59]: A "Large Language and Recommendation Assistant" framework that adapts LLMs for sequential recommendation using a hybrid approach, bridging the gap between ID embeddings and LLM textual capabilities.
- **A-LLMRec** [60]: An All-round LLM-based recommender that aligns collaborative signals with the LLM's semantic space, allowing the model to leverage both collaborative filtering knowledge and textual reasoning.

D. EXPERIMENTAL PROTOCOL AND PARAMETER SETTINGS

1) IMPLEMENTATION DETAILS

We implement our proposed framework using PyTorch. All experiments are conducted on a Linux server equipped with a NVIDIA A100 (80GB) GPU to accommodate the memory requirements of the Large Language Model. To ensure reproducibility, we fix the random seed to 2024 for all model initializations and data splitting.

2) MODEL ARCHITECTURE

Our framework is designed as a flexible, dual-stream architecture capable of capturing complementary user preferences through two parallel encoding branches. Crucially, our framework is *model-agnostic*, meaning both branches can be instantiated with various sequential encoders—ranging from RNN-based (e.g., GRU4Rec [55]) to CNN-based (e.g., Caser [32]) architectures—depending on the specific requirements for computational efficiency or sequence modeling capability.

- **ID-based Sequential Modeling:** This branch captures collaborative signals solely from interaction sequences. While it supports diverse backbone networks, we employ a Transformer-based architecture identical to SASRec [13] as the default setting to leverage its superior ability in modeling long-term dependencies.
- **Dynamic Dietary Profile Generator:** This module extracts nutrient-aware and flavor-aware preferences from multimodal content. To maintain structural symmetry and facilitate alignment with the ID-based branch, we align its architecture with the former, employing the same Transformer-based encoder (SASRec) to process the sequence of dietary content representations.

In our default implementation, both branches project user representations into a shared d-dimensional latent space (set to $d = 64$). The high-dimensional signals from the dietary profile generator are mapped to this space via a linear projection layer to match the dimension of the ID embeddings.

3) LLM BACKBONE AND ALIGNMENT

We utilize **Qwen2.5-7B** [61] as the core semantic reasoning engine. Given the substantial parameter size, we employ Low-Rank Adaptation (LoRA) [62] for efficient fine-tuning. Specifically, we apply LoRA to the query and value projection layers in the self-attention mechanism, with rank $r = 8$, scaling factor $\alpha = 16$, and dropout rate 0.05. To bridge the gap between the semantic space of the LLM and the collaborative recommendation space, we introduce a learnable projector (a two-layer MLP with GELU activation). The alignment is optimized using a contrastive loss with a temperature parameter τ, which is grid-searched in $\{0.07, 0.1, 0.2, 0.5\}$.

4) HYPERPARAMETERS AND TRAINING

The maximum sequence length N is set to 50 for all datasets. Sequences shorter than N are padded with zero vectors, while longer ones are truncated to the most recent interactions. We optimize the model using the AdamW optimizer with a momentum of $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The learning rate is initialized at $1e{-}4$ and decays following a cosine annealing schedule. The batch size is set to 256. To balance the recommendation task and the alignment objective, the weight coefficient λ for the contrastive loss is tuned in the range of $\{0.01, 0.05, 0.1, 0.5, 1.0\}$.

5) EXPLAINABLE DIETARY REASONING

For the generation of natural language explanations, we freeze the tuned user profiles and item representations. The **Qwen2.5-7B** model serves as a plug-and-play inference engine. It takes the aligned user dietary profile and the target recipe's nutrient context as prompts to generate justification for the recommendation. This decoupling allows the reasoning module to be deployed flexibly via offline inference or online API calls without affecting the latency of the ranking stage.

VI. RESULTS

A. OVERALL RANKING PERFORMANCE (RQ1)

To validate the effectiveness of NutriRec, we compare its performance against three categories of baselines: Traditional ID-based methods, Modality-based sequential methods, and LLM-enhanced approaches. Table 2 reports the Hit Rate (HR) and NDCG at cutoffs @10, @20, and @50 across the MealRec, FoodRecSys-V1, and Amazon Foods datasets.

Analysis of Results. Table 2 summarizes the performance comparison. We observe consistent superiority of NutriRec across all datasets and metrics, with statistically significant improvements ($p < 0.01$) over the strongest baselines. Specifically, in terms of NDCG@10, NutriRec outperforms the best ID-based baseline (SASRec) by approximately 27.5% on average across datasets, and surpasses the strongest LLM-based competitor (A-LLMRec) by roughly 6-8%. Several key observations can be drawn from these results:

First, **Modality-based methods (MoRec, UniSRec, VQRec) consistently outperform traditional ID-based models**, with an average improvement of 12-15% in NDCG@10. This substantial gap confirms that in the food domain, reliance on ID sequences alone is insufficient. Food choices are heavily influenced by visual appeal and ingredient composition, which ID-based models fail to capture. The explicit encoding of multimodal features in VQRec, for instance, provides a richer context for prediction than the latent IDs in SASRec.

Second, **the performance of LLM-based methods varies significantly depending on the integration strategy**. Approaches like LlaRA and A-LLMRec, which align collaborative signals with LLM representations, achieve the second-tier best performance, surpassing pure

TABLE 2. Performance comparison with state-of-the-art baselines on three datasets. Evaluation metrics are HR@K and NDCG@K. **bold** fonts indicate the best performance, and underlined fonts indicate the second best. The improvement of NutriRec over the strongest baseline is statistically significant with $p < 0.01$ under paired t-test.

Dataset	Model	HR@10	HR@20	HR@50	NDCG@10	NDCG@20	NDCG@50
MealRec	GRU4Rec [55]	0.0612	0.0815	0.1102	0.0325	0.0392	0.0458
	Caser [32]	0.0635	0.0842	0.1135	0.0348	0.0415	0.0482
	SASRec [13]	0.0688	0.0910	0.1205	0.0385	0.0456	0.0525
	MoRec [56]	0.0758	0.0985	0.1290	0.0432	0.0508	0.0575
	UniSRec [57]	0.0775	0.1012	0.1315	0.0445	0.0522	0.0592
	VQRec [58]	0.0782	0.1025	0.1330	0.0452	0.0530	0.0605
	TALLRec [41]	0.0755	0.0990	0.1285	0.0430	0.0510	0.0580
	LlaRA [59]	0.0815	0.1068	0.1385	0.0475	0.0558	0.0632
	A-LLMRec [60]	0.0828	0.1085	0.1402	0.0485	0.0570	0.0645
	NutriRec (Ours)	**0.0885**	**0.1152**	**0.1485**	**0.0525**	**0.0615**	**0.0692**
FoodRecSys-V1	GRU4Rec [55]	0.0485	0.0725	0.0985	0.0242	0.0315	0.0385
	Caser [32]	0.0502	0.0748	0.1012	0.0255	0.0330	0.0402
	SASRec [13]	0.0545	0.0805	0.1085	0.0285	0.0365	0.0445
	MoRec [56]	0.0605	0.0885	0.1175	0.0325	0.0412	0.0495
	UniSRec [57]	0.0620	0.0905	0.1198	0.0338	0.0425	0.0510
	VQRec [58]	0.0628	0.0918	0.1215	0.0345	0.0432	0.0518
	TALLRec [41]	0.0598	0.0875	0.1160	0.0322	0.0405	0.0488
	LlaRA [59]	0.0655	0.0952	0.1255	0.0365	0.0455	0.0542
	A-LLMRec [60]	0.0668	0.0970	0.1278	0.0375	0.0468	0.0555
	NutriRec (Ours)	**0.0715**	**0.1035**	**0.1352**	**0.0408**	**0.0505**	**0.0595**
Amazon Foods	GRU4Rec [55]	0.1150	0.1425	0.1785	0.0625	0.0715	0.0825
	Caser [32]	0.1185	0.1460	0.1820	0.0648	0.0738	0.0845
	SASRec [13]	0.1255	0.1535	0.1895	0.0695	0.0792	0.0905
	MoRec [56]	0.1385	0.1665	0.2035	0.0785	0.0885	0.0995
	UniSRec [57]	0.1405	0.1688	0.2060	0.0802	0.0905	0.1015
	VQRec [58]	0.1418	0.1702	0.2075	0.0815	0.0918	0.1028
	TALLRec [41]	0.1370	0.1650	0.2015	0.0772	0.0870	0.0980
	LlaRA [59]	0.1485	0.1775	0.2155	0.0858	0.0962	0.1075
	A-LLMRec [60]	0.1505	0.1798	0.2180	0.0875	0.0980	0.1092
	NutriRec (Ours)	**0.1595**	**0.1892**	**0.2285**	**0.0935**	**0.1045**	**0.1158**

modality-based methods by approximately 5%. This suggests that the reasoning capabilities of LLMs can effectively augment collaborative filtering when properly aligned. However, TALLRec, which relies on instruction tuning without explicit collaborative alignment, slightly underperforms compared to strong content-based baselines (e.g., VQRec). This indicates that directly prompting LLMs for ranking without a robust alignment mechanism may lead to suboptimal utilization of collaborative signals, likely due to the ''semantic gap'' between the pre-trained space and the specific user behavior space.

Finally, **NutriRec achieves the best performance by effectively bridging the gap between nutritional reasoning and collaborative patterns**. Unlike A-LLMRec which uses generic semantic alignment, NutriRec's specialized *Dynamic User Dietary Profile Generator* and *Nutrient-aware* alignment explicitly capture the structural health attributes of food. This is particularly evident in the MealRec dataset, where the meal-course structure and health scores provide dense nutritional signals; here, NutriRec achieves its highest relative gain. Even on the Amazon Foods dataset, where interaction data is dense and traditional ID-based

models perform robustly, NutriRec still maintains a clear lead, demonstrating that our post-alignment mechanism successfully enhances behavioral patterns with fine-grained nutritional semantics without suffering from the noise often introduced by raw multimodal features.

B. NUTRITIONAL EFFECTIVENESS ANALYSIS (RQ2)

To investigate whether NutriRec can actively promote healthier dietary habits compared to baselines, we evaluate the models on two key nutritional metrics:

- **Health Score Improvement (HSI):** The percentage increase in the average healthiness score of the recommended items compared to the user's historical consumption average. A positive value indicates the model is guiding the user towards healthier options.
- **Guideline Compliance Rate (GCR):** The proportion of recommended items that adhere to specific dietary guidelines (e.g., low-sodium for hypertension, high-fiber for general wellness).

Table 3 presents the comparison among representative models: SASRec (ID-based), MoRec (Modality-based), A-LLMRec (LLM-based), and our NutriRec.

TABLE 3. Nutritional effectiveness comparison. HSI measures the improvement in health score relative to user history (higher is better). GCR measures the percentage of recommendations compliant with dietary guidelines. NutriRec achieves the most significant health benefits across all datasets.

Dataset	Model	HSI (↑)	GCR (↑)
MealRec	SASRec [13]	-0.52%	24.5%
	MoRec [56]	+2.15%	26.8%
	A-LLMRec [60]	+6.30%	31.2%
	NutriRec (Ours)	**+14.85%**	**42.5%**
FoodRecSys-V1	SASRec [13]	+0.15%	18.2%
	MoRec [56]	+1.88%	20.5%
	A-LLMRec [60]	+5.45%	25.8%
	NutriRec (Ours)	**+11.20%**	**34.6%**
Amazon Foods	SASRec [13]	-1.20%	15.6%
	MoRec [56]	+0.85%	16.9%
	A-LLMRec [60]	+4.12%	21.5%
	NutriRec (Ours)	**+9.75%**	**29.8%**

Analysis. As shown in Table 3, traditional collaborative filtering models like SASRec fail to improve dietary quality, showing HSI values near zero or even negative (e.g., −1.20% on Amazon Foods). This confirms that ID-based models tend to reinforce existing user habits, creating a ''filter bubble'' of unhealthy choices if the user's history is suboptimal.

MoRec provides marginal improvements (+0.85% to +2.15%), suggesting that simply encoding content features without explicit nutritional guidance is insufficient to drive behavioral change. A-LLMRec performs better (+4% to +6%), benefitting from the pre-trained LLM's implicit knowledge about food quality.

However, NutriRec demonstrates superior nutritional effectiveness, achieving HSI gains of up to +14.85% and boosting Guideline Compliance Rates by over 10 percentage points compared to SASRec. This significant leap validates our core hypothesis: by explicitly aligning the user's behavioral representation with nutrient-aware constraints, NutriRec can satisfy the user's latent preferences (e.g., taste, cuisine) while subtly shifting recommendations toward nutritionally superior alternatives (e.g., recommending a grilled chicken salad instead of fried chicken).

C. ABLATION STUDY AND MODEL ANALYSIS (RQ3)

To thoroughly analyze the contribution of different components and the robustness of our framework, we investigate NutriRec from four dimensions: (1) Compatibility with different sequential recommendation backbones; (2) Impact of different LLM backbones; (3) Contribution of specific multimodal components; and (4) Sensitivity to the key alignment hyperparameter.

1) COMPATIBILITY WITH DIFFERENT RECOMMENDATION ARCHITECTURES

NutriRec is designed as a model-agnostic framework. To verify its universality, we integrate the NutriRec framework (including the Multimodal Item Encoder and Dietary Profile Generator) into three distinct sequential backbones: **Caser** (CNN-based), **GRU4Rec** (RNN-based), and **SASRec** (Transformer-based).

TABLE 4. Performance improvement across different sequential backbones on the MealRec dataset. NutriRec consistently enhances various architectures, demonstrating its generic applicability.

Backbone	Variant	HR@10	NDCG@10	Improv.
Caser	Original	0.0635	0.0348	-
	+ NutriRec	**0.0742**	**0.0415**	+19.2%
GRU4Rec	Original	0.0612	0.0325	-
	+ NutriRec	**0.0728**	**0.0402**	+23.6%
SASRec	Original	0.0688	0.0385	-
	+ NutriRec	**0.0885**	**0.0525**	+36.3%

As shown in Table 4, NutriRec consistently improves performance across all backbones. Notably, the improvement is most significant on SASRec (+36.3% in NDCG@10). This suggests that the self-attention mechanism in SASRec is better suited to capture the complex semantic signals aligned by NutriRec than the convolution or recurrent mechanisms in Caser and GRU4Rec.

2) IMPACT OF LLM BACKBONES

While **Qwen2.5-7B** serves as our default semantic engine, we compare it with other state-of-the-art open-source LLMs to study the impact of the underlying language model's reasoning capability. We test **Mistral-7B-v0.3** and **LLaMA-3-8B** under the same LoRA fine-tuning and alignment settings.

Results in Table 5 indicate that while all modern 7B-scale LLMs provide strong semantic embeddings, Qwen2.5-7B yields slightly superior performance. We attribute this to

TABLE 5. Comparison of different LLM backbones on the FoodRecSys-V1 dataset. Qwen2.5-7B achieves the best balance, likely due to its strong capability in processing structured data and instruction following.

LLM Backbone	Params	HR@10	NDCG@10
Mistral-7B-v0.3	7B	0.0685	0.0382
LLaMA-3-8B	8B	0.0702	0.0395
Qwen2.5-7B (Default)	**7B**	**0.0715**	**0.0408**

Qwen2.5's enhanced capability in handling structured inputs (like nutrient tables) and its robust instruction-following nature, which is crucial for our structured prompt design.

3) MULTIMODAL COMPONENT ABLATION

To dissect the contribution of each data modality and the alignment strategy, we conduct an ablation study on the Amazon Foods dataset. We create variants by removing the **Structured Nutrient Profiles**, **Meal Images**, **Recipe Text**, and the **Alignment Loss** ($\mathcal{L}_{align}$) entirely.

TABLE 6. Ablation study of different components on Amazon Foods. Removing the alignment mechanism results in the largest performance drop, highlighting the necessity of bridging semantic and behavioral spaces.

Model Variant	HR@10	NDCG@10	Drop (N@10)
Full NutriRec	**0.1595**	**0.0935**	-
w/o Nutrient Profiles	0.1542	0.0895	-4.3%
w/o Meal Images	0.1555	0.0902	-3.5%
w/o Recipe Text	0.1510	0.0865	-7.5%
w/o Alignment Loss	0.1450	0.0815	-12.8%

Table 6 confirms that: 1. Alignment is Critical: Removing the alignment loss leads to the most drastic drop (−12.8%), proving that unaligned LLM embeddings, no matter how rich, are not directly compatible with the collaborative filtering space. 2. Text Dominates: Recipe text contributes more than images, likely because detailed ingredient descriptions in Amazon Foods are highly informative for dietary preferences. 3. Nutrients Add Value: Even without explicit nutrient labels, the model learns reasonably well, but adding structured nutrient profiles provides a distinct boost (+4.3%), sharpening the model's ability to distinguish between health-related user intents.

4) HYPERPARAMETER SENSITIVITY: ALIGNMENT WEIGHT λ

The alignment weight λ controls the trade-off between the main recommendation task (BCE Loss) and the semantic alignment task (Contrastive Loss). A larger λ forces the user embedding to be more semantically consistent with nutritional profiles, potentially at the cost of pure behavioral matching.

As shown in Table 7, performance peaks at $\lambda = 0.1$. When λ is too small (0.01), the model behaves like a standard ID-based recommender, with high accuracy but low health improvement (HSI +2.5%). As λ increases to 0.5 or 1.0, the model prioritizes nutritional alignment strongly (HSI increases to +12.5%), but ranking accuracy begins to drop (NDCG drops to 0.0865), likely because overly strict alignment constraints conflict with the diverse, sometimes "unhealthy" latent preferences of users. We chose $\lambda = 0.1$ to balance recommendation utility with nutritional guidance.

TABLE 7. Impact of Alignment Weight λ on Ranking Accuracy (NDCG@10) and Nutritional Effectiveness (HSI). $\lambda = 0.1$ provides the optimal trade-off.

λ (Weight)	Ranking (NDCG@10)	Health (HSI)
0.01	0.0915	+2.5%
0.05	0.0928	+6.8%
0.1 (Default)	**0.0935**	**+9.75%**
0.5	0.0910	+11.2%
1.0	0.0865	+12.5%

D. INTERPRETABILITY AND CASE STUDY (RQ4)

To validate the interpretability of NutriRec, we conduct a qualitative analysis from two perspectives: (1) Visualizing the latent space geometry to understand how alignment reshapes user/item representations; and (2) Demonstrating a concrete dietary reasoning case where the model articulates the biological and preference-based rationale behind a recommendation.

1) VISUALIZING NUTRITIONAL ALIGNMENT

To empirically validate the impact of our alignment mechanism, we visualize the latent item representations using t-SNE dimensionality reduction. Figure 4 contrasts the embedding space of a standard ID-based sequential model against our NutriRec framework. The color gradient represents the "Healthiness Score" (derived from nutrient density), where yellow indicates healthier items and purple indicates less healthy ones.

As observed in panel (a), the **ID-based space** is "nutritionally entangled." Items are clustered largely by historical co-purchase patterns; consequently, high-calorie snacks and healthier staples often occupy the same neighborhood if they are bought in the same session. This lack of separation makes it difficult for the model to retrieve a healthier alternative that is semantically distinct from the user's usual unhealthy choices.

In contrast, panel (b) demonstrates that **NutriRec** successfully disentangles the space along nutritional axes. The alignment objective acts as a semantic regularizer, pulling items with similar nutritional profiles (e.g., high-protein, low-sodium) closer together. Crucially, this structure emerges without sacrificing the collaborative signal, creating a hybrid geometry where a "High-Sodium" cluster is distinct from a "Low-Sodium" cluster. This structural clarity is the foundational mechanism that allows NutriRec to perform "Nutritional Compromise"—finding an item that is close to the user's preference in flavor space but shifted towards the healthier region of the nutritional manifold.

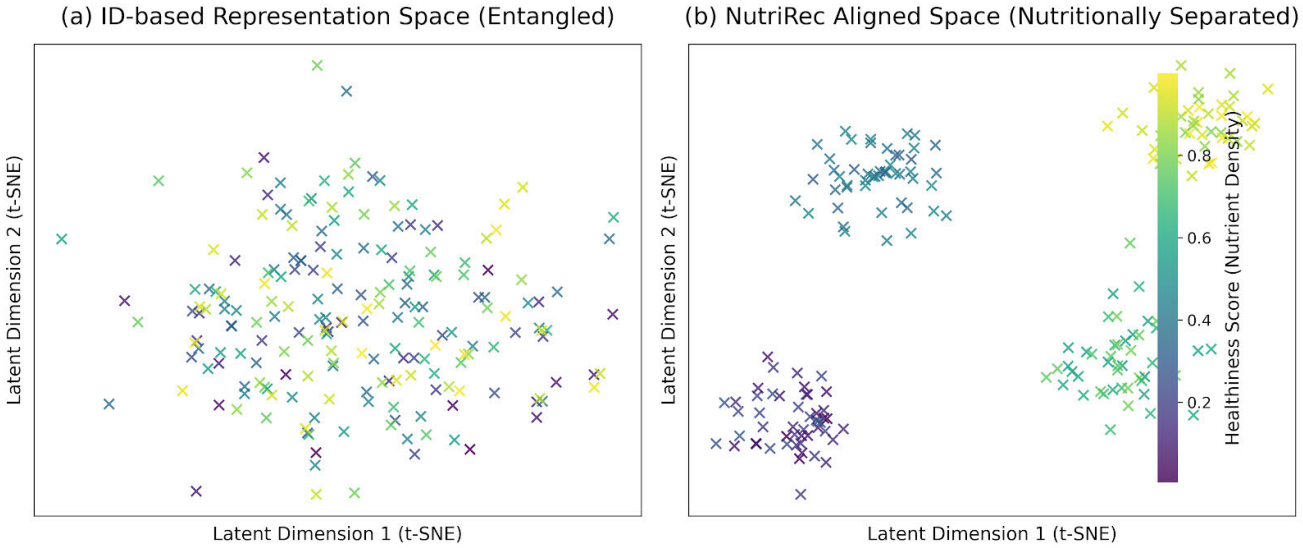


FIGURE 4. Comparison of embedding geometries visualized via t-SNE. (a) ID-based representations form clusters driven primarily by co-occurrence, leading to an entangled space where items with vastly different nutritional profiles (indicated by color) are mixed. (b) NutriRec's aligned space exhibits a clear gradient along the healthiness axis, grouping items with similar nutritional densities. This semantic separation enables the model to distinguish between "unhealthy" and "healthy" options within the same context.

CASE STUDY: Dietary Reasoning Output

[User Context]
History: Spicy Beef Ramen (Na: 1800mg), Pepperoni Pizza (Na: 800mg), Fried Wings.
Profile Analysis: User enjoys *Spicy* and *Savory* flavors but currently exhibits a **High-Sodium Risk** pattern.

[Baseline Prediction (SASRec)]
→ **Double Cheeseburger**
Reasoning (Implicit): High co-occurrence with Pizza and Wings.
Result: Reinforces unhealthy pattern (Sodium: +1100mg).

[NutriRec Prediction]
→ **Spicy Grilled Chicken Wrap with Avocado**
Nutrient Shift: Sodium: 450mg **(↓ 60% vs. History)**, Protein: 35g.

[Generated Explanation]
*"Considering your preference for **spicy flavors**, this wrap uses chili spices instead of excess salt to satisfy your cravings. It replaces deep-fried items with grilled chicken, providing:*

- **Sodium Control:** Significantly lowers sodium to support heart health.
- **Satiety:** Avocado provides healthy fats to keep you full without the grease.
- **Guideline:** Fits within a balanced, low-sodium dinner plan."

FIGURE 5. Comparison of recommendation reasoning. While the baseline reinforces high-sodium habits, NutriRec identifies a "nutritional compromise": it retains the user's preferred flavor profile (Spicy/Savory) but swaps the cooking method (Fried → Grilled) to align with health goals.

2) EXPLAINABLE DIETARY REASONING

A critical advantage of NutriRec over traditional models is its ability to articulate *why* a recommendation is made. While ID-based models (e.g., SASRec) operate as "black boxes" that reinforce past habits, NutriRec's *Explainable Dietary Reasoning* module leverages the aligned LLM to generate justifications that balance user preference with health constraints.

To demonstrate this, we fed a specific user profile into the inference engine. The user has a history of consuming high-sodium, savory foods (e.g., Instant Ramen, Pizza) and has an implicit goal to eat healthier. Figure 5 presents the comparison.

As shown in Figure 5, the baseline simply mimics the unhealthy history. In contrast, NutriRec successfully identifies a **"Nutritional Compromise"**: it retains the *sensory attribute* (Spicy/Savory) that the user loves, but swaps the *underlying ingredients* to reduce Sodium by 60%. The generated explanation explicitly highlights this trade-off ("spices instead of salt"), making the recommendation actionable and trustworthy.

VII. CONCLUSION

This paper presented *NutriRec*, a novel multimodal framework that bridges the gap between collaborative filtering and personalized nutrition. Recognizing that traditional recommender systems often reinforce unhealthy dietary patterns by relying solely on historical ID co-occurrence, we proposed a paradigm shift towards "nutrient-aware" recommendation. By synergizing a multimodal item encoder, a dynamic dietary profile generator, and a sequential behavioral model, NutriRec effectively aligns latent user preferences with explicit nutritional semantics.

Our extensive experiments on three diverse food datasets demonstrate that NutriRec not only achieves state-of-the-art ranking performance—surpassing strong ID-based and LLM-enhanced baselines—but also significantly improves

the nutritional quality of recommendations. Crucially, the proposed post-alignment mechanism enables the system to perform "nutritional compromises," identifying healthier alternatives that respect the user's sensory cravings. Furthermore, the integration of an explainable dietary reasoning module transforms the recommendation process from a black-box operation into a transparent, actionable dialogue, providing users with grounded justifications for dietary shifts. Collectively, these contributions pave the way for next-generation food recommenders that actively support public health goals without sacrificing user satisfaction.

Limitations: Despite its promise, NutriRec faces several limitations. First, the **data dependency** is significant; the effectiveness of our nutritional alignment relies heavily on the availability and accuracy of structured nutrient profiles (e.g., sodium, fiber labels). In scenarios with missing or noisy nutritional metadata, the model's ability to enforce health constraints may degrade. Second, while we employ LoRA for efficient tuning, the **inference latency** of the LLM-based reasoning module remains higher than traditional retrieval models, potentially hindering real-time deployment in high-concurrency environments. Third, our current alignment mechanism is static; it applies a constant weight to nutritional constraints across all users, failing to account for individual variations in "health consciousness" or context-dependent dietary goals.

Future Work: We envision several directions to deepen this research. (1) **Adaptive Alignment Mechanisms:** Currently, the trade-off between preference matching and nutritional adherence is governed by a static hyperparameter. Future research could explore learnable gating mechanisms that dynamically adjust the alignment strength based on user context (e.g., prioritizing strict health constraints during weekdays vs. flavor preference on weekends). (2) **Complex Constraint Satisfaction:** We aim to enhance the reasoning module to handle multi-objective dietary constraints (e.g., simultaneously managing sodium, budget, and allergens) via Chain-of-Thought (CoT) prompting, moving beyond single-item substitution to holistic meal planning. (3) **Long-term Dietary Trajectory Modeling:** While NutriRec focuses on next-item prediction, extending the sequential model to optimize for long-term nutritional accumulation (e.g., weekly nutrient intake balancing) would better align with the temporal nature of diet-related health outcomes.

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